

A Representative Sampling Method for Peer Encouragement Designs in Network Experiments

Yanyan Li,^a Sha Yang,^b Qing Liu^c

^a Department of Marketing & Entrepreneurship, C. T. Bauer College of Business, University of Houston, Houston, Texas 77004. yli171@bauer.uh.edu.

^b Department of Marketing, Marshall School of Business, University of Southern California, Los Angeles, California 90007. shayang@marshall.usc.edu.

^c Department of Marketing, Wisconsin School of Business, University of Wisconsin-Madison, Madison, Wisconsin 53706. qliu@bus.wisc.edu.

Abstract

Network experiments using peer encouragement designs are widely used to estimate the direct and indirect effects of social marketing campaigns. To satisfy the Stable Unit Treatment Value Assumption (SUTVA), contaminated nodes are typically excluded from analysis. This exclusion approach creates two problems: underrepresentation, where post-exclusion samples are not representative of the population, and undersupply, where post-exclusion samples are too small for reliable inference. We propose a representative ego network sampling method that directly addresses these challenges. The method embeds a distance constraint within a Metropolis-Hastings algorithm to generate contamination-free ego network samples whose joint distribution of key network attributes converges to that of the population. Using both simulated and real-world networks, we show that the proposed method consistently generates larger and more representative samples than the standard exclusion approach. These samples produce more accurate and reliable estimates of average and heterogeneous treatment effects and improve statistical inference. The advantages are especially pronounced when experiments require large samples, population networks are small or dense, or higher-order contamination must be controlled. Practically, the method can be adapted for various applications, providing a valuable tool for firms designing and evaluating social marketing campaigns.

Keywords: peer encouragement designs; network experiments; interference; representative sampling; metropolis-hastings; social influence

1. Introduction

With the continued growth of social media, social marketing campaigns such as referral campaigns and social advertising have become increasingly common. The effectiveness of these campaigns depends not only on the direct responses of targeted individuals but also on the influence they exert on their social connections. To maximize return on marketing investment, companies need to understand both direct and indirect effects when selecting the targeting group. Pilot studies can inform this decision by identifying which subpopulations are more responsive, how strongly these subpopulations influence their social connections, and how expected returns vary across alternative targeting interventions.

Peer encouragement designs are widely used to estimate the direct and indirect effects of social marketing campaigns (Bapna and Umyarov 2015, Eckles et al. 2016, Ascarza et al. 2017). In these designs, ego networks, each consisting of a focal individual (ego) and the individuals connected to the ego (alters), are randomly sampled and assigned to treatment or control conditions (Figure 1). Treatment is implemented at the ego level. The direct treatment effect is inferred from outcome differences between treated and control egos, and the indirect treatment effect is inferred from outcome differences among their connected alters (Aral 2016).

To satisfy the Stable Unit Treatment Value Assumption (SUTVA), contaminated nodes are typically excluded from analysis. Contamination occurs when treated and control egos are directly connected or when multiple egos are indirectly connected through a common alter (Figure 2). This exclusion approach creates two problems. The first is underrepresentation: because high-degree egos (i.e., those with many connections) are more likely to be contaminated and excluded, the remaining egos tend to be more homogeneous and less representative of the population. When treatment effects vary with network position, estimates and experimental findings based on such nonrepresentative samples may not generalize to the population (Bapna and Umyarov 2015). The second is undersupply:

the remaining sample may be too small to deliver adequate statistical precision or power for estimating average and heterogeneous treatment effects. To compensate, researchers have to begin with a much larger initial sample, which is inefficient and costly when treatments involve monetary incentives (Feit and Berman 2019).

In this research, we propose a representative ego network sampling method that directly addresses the problems of underrepresentation and undersupply. The proposed method has three key advantages over the traditional exclusion approach. First, it generates treatment and control samples that represent the population network well, such that the joint distribution of key network attributes among sampled egos converges to that of the population. These representative samples improve the estimation of average and heterogeneous treatment effects and enhance the generalizability of experimental findings. Second, the method proactively controls contamination by embedding a distance constraint in the sampling procedure, allowing researchers to specify the degree of contamination to be controlled for. Third, the desired sample size can be pre-specified in the method, making it possible to obtain larger valid samples for causal inference. Taken together, the proposed method provides a powerful and efficient way to draw representative, large, and uncontaminated ego network samples for causal inference in network experiments using peer encouragement designs.

We conduct simulations on both simulated and real-world networks to compare samples generated by the proposed method (hereafter “representative samples”) with samples generated by the exclusion approach (hereafter “post-exclusion samples”). Our method employs a constrained Metropolis-Hastings algorithm that minimizes the difference between the samples and population distributions in degree and clustering coefficient, two core network attributes that may affect network structure and treatment effect heterogeneity. Compared with post-exclusion samples, the representative samples better represent the population not only on these optimization variables but also on other network

attributes and individual characteristics correlated with them. For variables unrelated to the optimization variables, representativeness is comparable across the two methods.

We further examine the implications for causal inference. We consider a network experiment with a peer encouragement design for evaluating a targeted intervention, in which treated egos respond to the intervention and influence alters heterogeneously according to their network attributes. Representative samples produce more accurate and reliable estimates of both average and heterogeneous treatment effects. Across all networks, representative samples consistently achieve lower mean absolute error (MAE) and root mean squared error (RMSE), along with higher statistical power and confidence interval coverage, than post-exclusion samples. These advantages are especially pronounced when experiments require large sample sizes, when the population network is small or dense, or when both first- and second-degree contamination must be controlled. Under these conditions, the underrepresentation and undersupply problems from the exclusion approach are more severe, making the benefits of the proposed method especially meaningful.

This research contributes in three ways. First, it introduces an efficient method for generating representative, large, and uncontaminated ego network samples in network experiments using peer encouragement designs. Such designs have been widely adopted in a variety of marketing contexts, including viral marketing (Hinz et al. 2011), music subscription (Bapna and Umyarov 2015), user feedback (Eckles et al. 2016), CRM campaigns (Ascarza et al. 2017), and churn management (Godinho de Matos et al. 2018). However, how to generate representative ego network samples while controlling treatment contamination has been identified as a “methodological limitation” (Bapna and Umyarov 2015). Our research directly addresses this challenge. Through extensive simulations on both simulated and real-world networks, we show that underrepresentation and undersupply are substantial concerns under the exclusion approach and that our method effectively mitigates both problems.

Second, our research provides boundary conditions and guidelines on when these challenges are most likely to arise. The problems of underrepresentation and undersupply become more severe when experiments require large sample sizes, when the population network is small or dense, or when both first- and second-degree contamination must be controlled. These conditions are common in practice. For instance, for low-probability events such as ad clicks or conversions, large samples may be required to achieve sufficient statistical power to detect treatment effects, increasing the risk of sample contamination. Moreover, even large platforms such as Facebook and LinkedIn often conduct experiments on smaller local subnetworks. Our method provides a practical solution in these settings.

Third, our benchmark analysis shows that representative samples produce more accurate and reliable estimates of both average and heterogeneous treatment effects and improve statistical inference. These improvements have two managerial implications. First, representative samples help companies more accurately identify whom to target. This is particularly important for interventions such as referral campaigns or social advertising, where companies must consider not only who is likely to respond but also who is likely to influence others. Second, the larger valid sample sizes generated by our method are especially valuable when the outcomes of interest are rare. Our method helps companies obtain more precise and generalizable experimental evidence for targeting and resource-allocation decisions.

The remainder of the article is organized as follows. Section 2 reviews peer encouragement designs and highlights the methodological challenges. Section 3 presents the proposed representative ego network sampling method. Section 4 evaluates the quality of representative samples relative to post-exclusion samples. Section 5 demonstrates the advantages of representative samples for estimating average and heterogeneous treatment effects. Section 6 concludes with implications and potential extensions.

2. Peer Encouragement Designs for Network Experiments

2.1 Peer Encouragement Designs and Causal Inference

Peer encouragement designs are widely used in network experiments to estimate the direct treatment effect of a marketing intervention on targeted individuals (egos) and the indirect treatment effect on those connected to them (alters) (Aral 2016). Under this design, egos are randomly assigned either to the treatment condition ($d_i = 1$), in which they receive the intervention intended to change their behavior, or to the control condition ($d_i = 0$). Although alters do not receive the treatment directly, their outcomes may be indirectly influenced when the connected ego is treated. Figure 1 illustrates the peer encouragement design and how the direct and indirect treatment effects are identified.

== Insert Figure 1 About Here ==

Let $Y_i(d_i, d_{n(i)})$ denote the potential outcome of individual i as a function of i 's own treatment status d_i and the treatment status of its immediate connections $d_{n(i)}$, where $n(i)$ denotes all immediate neighbors of i . The direct treatment effect (DTE) is the effect of treatment on treated egos, identified by comparing the outcomes of egos assigned to treatment with those assigned to control when none of their neighbors is treated:

$$\tau_i = Y_i(d_i = 1, d_{n(i)} = 0) - Y_i(d_i = 0, d_{n(i)} = 0). \quad (1)$$

The indirect treatment effect (ITE) is the effect on untreated individuals who are connected to treated egos. For alter j of ego i , $d_j = 0$ by design, but alter j may be indirectly exposed to treatment through ego i , so that $d_{n(j)} = d_i$. Specifically, $d_{n(j)} = 1$ when alter j is linked to a treated ego and $d_{n(j)} = 0$ when linked to a control ego. The ITE is identified by comparing outcomes of alters linked to treated egos with those linked to control egos:

$$\gamma_j = Y_j(d_j = 0, d_{n(j)} = 1) - Y_j(d_j = 0, d_{n(j)} = 0). \quad (2)$$

2.2 Treatment Contamination and the Standard Exclusion Approach

Peer encouragement designs, like many other network experimental designs, are prone to treatment contamination. First-degree contamination occurs when two sampled egos are directly connected, and second-degree contamination occurs when two sampled egos share one or more common alters (Figure 2). These forms of contamination violate SUTVA (Rubin 1980) and complicate causal identification. For example, if ego i has one or more treated neighbors, estimation of the DTE is biased because $d_{n(i)} \neq 0$. Similarly, if alter j is connected to multiple egos, estimation of the ITE is biased because the indirect treatment condition of alter j is no longer determined only by ego i , that is $d_{n(j)} \neq d_i$. In sum, samples used for causal inference must satisfy two conditions to avoid contamination. First, no two sampled egos, within or across treatment conditions, should be directly connected, supporting clean estimation of the DTE. Second, no alter should be linked to more than one sampled ego, within or across treatment conditions, supporting clean estimation of the ITE.¹

== Insert Figure 2 About Here ==

The most common approach to minimizing treatment contamination is to exclude contaminated samples from causal inference. Despite its simplicity, this approach faces two challenges. The first is underrepresentation: post-exclusion samples may not represent the population network. When treatment effects are heterogeneous across subpopulations, nonrepresentative samples can produce biased estimates and limit the generalizability of the findings. This concern is well documented. Bapna and Umyarov (2015) note that the excluded users “are not random since high-degree individuals are more likely to be friends of both groups” and describe this as a “difficult and not yet solved problem in network experiments.” Sun et al. (2021) acknowledge that excluding contaminated alters may reduce sample heterogeneity. Walker and Muchnik (2014) note that individuals receiving

¹ A related literature studies network interference under cluster randomization (Ugander et al. 2013, Eckles et al. 2017), where treatment is assigned at the cluster level. This literature focuses on population-level average treatment effect. In contrast, we focus on local social influence within ego networks. This difference motivates the two contamination-control conditions described here.

different types of indirect exposure to treatment, such as alters connected to multiple egos, may differ systematically from the broader population.

The second challenge is undersupply: excluding contaminated samples may leave too few qualified observations, reducing the precision and power of statistical inference (Aral and Walker 2014, Walker and Muchnik 2014). Moreover, inferences on heterogeneous treatment effects (HTE) can be challenging because certain subpopulations may be absent from the remaining samples. Understanding how different subpopulations respond to treatment is essential for developing effective targeted strategies. Prior research shows that individuals in different network positions may respond differently to interventions and exert different levels of social influence (Goldenberg et al. 2009, Trusov et al. 2010, Hinz et al. 2011, Iyengar et al. 2011). Individual characteristics, such as demographics, can also affect susceptibility and influence (Aral and Walker 2012, Aral and Dhillon 2018).

One may wonder how serious treatment contamination is in real-world field experiments. Through extensive simulation studies on both simulated and real-world networks, we find that sample size, population network size, and population network density strongly influence contamination rates, amplifying the challenges of obtaining representative and sizable samples in many circumstances. Moreover, when both first- and second-degree contamination are controlled, the majority of egos may become ineligible for analysis, rendering the exclusion approach impractical. These findings help explain why prior studies relying on large-scale, sparse population networks while ignoring second-degree contamination reported low contamination rates (e.g., less than 5%) (Bapna and Umyarov 2015, Ascarza et al. 2017, Sun et al. 2021). Under those conditions, the problems of underrepresentation and undersupply are less severe.

However, many real-world experiments are conducted on smaller, denser networks and require large samples with strict adherence to SUTVA. When treatments are costly or risky, companies often

run pilot studies on relatively small networks. In addition, as interest-based communities become increasingly prevalent, controlling only first-degree contamination may be insufficient, and requiring control for both first- and second-degree contamination further exacerbates the problems of underrepresentation and undersupply. Our proposed method directly addresses these challenges and is particularly beneficial when large samples are required, when the population network is small or dense, or when higher-order contamination must be controlled.

3. Representative Ego Network Sampling Method

3.1 Theoretical Framework

In peer encouragement designs, the sampling unit is the ego network, consisting of an ego and its alters. To illustrate the framework, we begin with the general case where a set of ego networks is sampled from the population. In the next section, we explain how the algorithm can be adapted to generate separate samples for each treatment condition. Let G denote the set of all individuals in the population network and S the set of n sampled egos. Because the set of alters $A(S)$ is determined by the choice of egos S , we focus on sampling S and treat $A(S)$ as implicitly defined. Let $f(S)$ represent the distribution of a network attribute (e.g., degree) for individuals in set S , and let Δ denote a distance measure between two distributions. Then $\Delta(f(S), f(G))$ quantifies the difference between the distributions of the network attribute in the sampled egos S and in the population G . Following the idea of representative sampling (Kruskal and Mosteller 1979, 1980), we obtain ego network samples that represent the population by selecting $S \subseteq G$ to minimize $\Delta(f(S), f(G))$.

To control treatment contamination, we impose a constraint on the shortest path between any two sampled egos, defined as the minimum number of edges connecting them. The shortest-path threshold h reflects the level of contamination the experiment aims to control. For instance, clean identification of the DTE requires that sampled egos are free of first-degree contamination. This condition can be

ensured by setting $h = 2$, which requires that no two egos i and i' in S are directly connected. The constrained optimization is:

$$\min_{S \subseteq G} \Delta(f(S), f(G)) \quad (3)$$

$$\text{subject to } \text{ShortestPath}(i, i') \geq h, \forall i, i' \in S, i \neq i'. \quad (4)$$

We highlight three points. First, the objective function in (3) can be defined over one or multiple variables capturing individual heterogeneity in responses to a targeting intervention. We focus on two network attributes. The first is degree, which is the total number of immediate connections an individual has. It reflects an individual's local influence and is closely related to responses to targeted interventions and social influence (Goldenberg et al. 2009, Hu and Van den Bulte 2014). The second is clustering coefficient, which measures the likelihood that an individual's neighbors are also connected to one another.² Incorporating clustering coefficients ensures that sampled ego networks represent the population in local connectivity and homophily structure.

Second, we measure the distance between the sample and population distributions using the Kolmogorov-Smirnov (KS) Statistics, defined as the maximum absolute difference between the cumulative distribution functions (CDFs) F_G of $f(G)$ and F_S of $f(S)$,

$$\Delta(f(S), f(G)) = \max_x |F_G(x) - F_S(x)|. \quad (5)$$

where x denotes all possible values of the network attributes. When multiple variables are included, the distance is computed based on the joint distribution of these variables. For example, for degree and clustering coefficient, we first obtain the joint distributions of these two variables in the

² The clustering coefficient C_i is calculated as the fraction of possible triangles (i.e., a group of friends of three) through the focal individual i , $C_i = 2T(i)/K_i(K_i - 1)$, where $T(i)$ is the number of triangles through individual i , and K_i is the individual degree.

population and the sample and then compute the KS distance by taking the maximum absolute difference between the corresponding joint CDFs.³

Finally, the shortest-path threshold h is chosen by the researcher to control the separation between sampled egos. In practice, researchers can increase the threshold value to further space sampled egos apart when higher-order contamination is a concern.

3.2 The Metropolis-Hastings Algorithm for Representative Ego Network Sampling

The key advantage of our method over the traditional exclusion approach is that it generates uncontaminated and representative samples through an explicit optimization procedure. However, minimizing the distance between the sample and population as defined in (3), subject to the contamination constraint in (4), is analytically intractable. To address this challenge, we employ the Metropolis-Hastings (MH) algorithm (Hübler et al. 2008). The core concept involves constructing a target distribution $\pi(S)$ of the desirable samples such that samples closer to the population are more likely to be selected and then using the MH algorithm to obtain this target distribution and the optimal samples. We define the target distribution to be inversely related to the distance between the sample and the population:

$$\pi(S) = \frac{1}{\Delta(f(S), f(G))^p}. \quad (6)$$

The exponent p is a positive scalar that controls the sharpness of the distribution. Following Hübler et al. (2008), we set $p = 10 \cdot (k/N) \log_{10} N$, where k is the total number of edges and N is the total number of nodes in the population network. This specification rewards samples that more closely match the population, preventing lower-quality samples from dominating the sampling process.

³ We also explored alternative distance measures such as the Kullback-Leibler (KL) divergence and Anderson-Darling distance, and found that the KS metric performs better in retaining high-degree individuals in the sample. See Web Appendix A for details.

In the MH algorithm, the chain is initiated from an arbitrary starting sample (i.e. a set of ego networks in our case), and a candidate sample is proposed at each iteration. The candidate sample is accepted with a probability that is theoretically derived to ensure the chain satisfies the Ergodic theorem. In simpler terms, the algorithm is designed so that any state can be reached from any other state within a finite number of transitions. This ensures that the Markov chain generated by the algorithm will converge to the desired target distribution. For a general introduction to the MH algorithm and its theoretical properties, see Chib and Greenberg (1995).

When generating proposals for the MH algorithm in our method, we leverage sequential states to accelerate convergence (Chen and Yang 2007, Musalem et al. 2008, Hübler et al. 2008). Specifically, at each iteration, we randomly remove one ego network from the current state S and replace it with another ego network drawn from the available space, generating a proposal sample S' . Let $q(S'|S)$ denote the proposal distribution for moving from state S to S' . Under this construction, the proposal distribution is symmetric, i.e., $q(S'|S) = q(S|S')$. The acceptance probability is therefore $a(S', S) = \min(1, \frac{\pi(S')}{\pi(S)})$ for accepting the candidate S' . This Markov chain satisfies the standard conditions for convergence and generates samples distributed according to $\pi(S)$, as shown in prior research (Chib and Greenberg 1995, Hübler et al. 2008).

3.3 Applying the Metropolis-Hastings Algorithm to Treatment Conditions

In peer encouragement designs, sampled ego networks must be assigned to different treatment conditions. When applied to such designs, the MH algorithm needs to be adapted to sample ego networks for each treatment condition and ensure samples are equally optimized across conditions.

Suppose an experiment involves two conditions $d \in \{T, C\}$ and each condition requires n ego networks, denoted by S_T and S_C . A simple approach is to draw a representative sample of size $2n$ and assign egos to treatment and control using randomization, stratification, blocking or matching. However, because the ego network samples still exhibit power-law distributions in network attributes,

these approaches may fail to achieve balance across conditions. Moreover, when multiple network attributes are considered, stratification, blocking, or matching becomes increasingly complex due to the higher dimensionality.

To ensure that samples across conditions are balanced and representative of the population network, we draw samples for both conditions simultaneously using the MH algorithm. We begin by selecting $2n$ initial ego networks that satisfy the contamination constraint and randomly assigning them to S_T and S_C , each of size n . We then apply the MH algorithm to iteratively update samples for one condition at a time, which involves the following two steps.

First, we select which condition to update. To ensure equal sample quality across conditions and improve efficiency, we draw an indicator $g \sim \text{Bernoulli}(p)$, with $p = \frac{\Delta(f(S_T), f(G))}{\sum_{d \in \{T, C\}} \Delta(f(S_d), f(G))}$, to determine whether S_T (if $g = 1$) or S_C (if $g = 0$) is selected to be updated. This approach updates the condition with a larger distance from the population more frequently, helping maintain balance across conditions.

Second, given the current sample for the selected condition (e.g., S_T), we propose a new sample S'_T by randomly removing one ego i and replacing it with a new ego i' drawn from the population subject to the contamination constraint. Let $S_{D_r} = \{S_T \cup S_C \setminus i\}$ denote the set of all egos in both conditions excluding the removed ego i . Under first-degree contamination control, the newly added ego i' must be at least two hops away from all egos in S_{D_r} ; that is, i' cannot be any ego in S_{D_r} or an alter of those egos. As shown in Web Appendix B, the proposal distribution of moving from the current sample state to the new sample state is symmetrical. Combined with the acceptance probability $a(S', S)$, this construction satisfies the detailed balance condition and ensures the convergence of the Markov chain to the target distribution $\pi(S)$.

As the Markov chain converges, the algorithm generates samples from the target distribution that closely match the population network in degree and clustering coefficient and maintain comparable quality across conditions. Following Hübler et al. (2008), we keep the sample with the smallest distance (i.e. the lowest KS statistic) to the population after convergence as the final sample for each condition.⁴ The full algorithm can be found in Web Appendix C.

4. Evaluating the Quality of Representative Samples

We evaluate the quality of representative samples relative to the post-exclusion samples. Using simulations on both simulated and real-world networks, we first demonstrate the extent of underrepresentation and undersupply under the exclusion approach and examine four factors that influence how pronounced these issues are: sample size, population network size, population network density, and the contamination criterion under SUTVA. We then compare the representativeness of samples generated by our method with that of post-exclusion samples.

4.1 Simulation Setup

We use a total of six networks as the population: three real-world social networks and three simulated power-law cluster networks. The three real-world networks are LastFM-Asia (a social network of Asian LastFM users for music sharing) (Rozemberczki and Sarkar 2020), Twitch-Gamer (a social network of Twitch gaming users) (Rozemberczki and Sarkar 2021), and Pokec (a social network in Slovakia for online dating) (Takac and Zabovsky 2012).⁵ We select these networks for three reasons. First, they provide complete network information, including nodes and edges. In contrast, most networks used in published marketing studies lack detailed information on individual nodes or links

⁴ Our method is computationally efficient. In the simulation settings investigated in Section 4, where 500 ego networks per treatment condition are sampled from a population network of 100,000 nodes, the average runtime for 200,000 iterations is approximately 3,470 seconds on a PC with 128 GB RAM and an Intel i9 processor (3.6 GHz, 18 cores).

⁵ The network datasets can be downloaded from SNAP Datasets (Leskovec and Krevl 2014).

and report only network-level metrics such as average degree. This makes it difficult to replicate the actual population network used in those studies. Even if we were to reconstruct one based on the limited available information, it is unclear whether differences in results are driven by methodological differences or by variations in the underlying networks. Second, they cover a variety of network conditions in the real world. LastFM-Asia is small and dense, Pokec is large and sparse, and Twitch-Gamer falls in between, allowing assessment of method performance across diverse conditions. Third, Pokec approximates the large, sparse networks typically used in prior marketing studies employing peer encouragement designs (e.g., Ascarza et al. 2017). This allows us to replicate findings from the traditional exclusion approach, which reports low contamination rates (e.g., less than 5%) under first-degree contamination only.

The three simulated population networks are generated using the power-law cluster model (Holme and Kim 2002), which is widely observed in real-world networks and used in prior research (Albert and Barabási 2002, Mislove et al. 2007, Chen et al. 2018). We construct a baseline network, referred to as PowerLaw-Cluster-Base, to represent an "average" network based on the diverse set of conditions observed in the three real-world networks. The other two simulated networks include a small network (PowerLaw-Cluster-Small) and a sparse network (PowerLaw-Cluster-Sparse), relative to the baseline.

Table 1 reports key characteristics of the six networks including population size, number of edges, average degree, average clustering coefficient, average eigenvector centrality, density, and average neighbor degree. These networks span a wide range of conditions studied in prior research. For example, network size ranges from 7,624 (small) to 1,632,803 (large), and density ranges from 0.00002 (sparse) to 0.001 (dense). For brevity, we refer to these networks as LastFM, Twitch, Pokec, Base, Small, and Sparse in figures and parenthetical references throughout.

== Insert Table 1 About Here ==

4.2 Issues of Undersupply and Underrepresentation in Post-Exclusion Samples

We demonstrate the significance of undersupply and underrepresentation that arise when contaminated units are removed under the standard exclusion approach. We consider two contamination criteria: (1) first-degree contamination, which occurs when two sampled egos are directly connected, and (2) first- and second-degree contamination, which occurs when two sampled egos are directly connected or share at least one common alter.

Undersupply. For each of the six population networks, we randomly sample n ego networks, where n increases from 100 to 1,000 in increments of 100. For each sample, we compute the percentage of contaminated egos under each criterion. This yields a total of 10 (sample sizes) $\times$ 2 (contamination criteria) $\times$ 6 (population networks) percentages. Figure 3 reports these percentages under first-degree and first- and second-degree contamination, respectively. Because contaminated egos are removed under the exclusion approach, these percentages indicate the severity of undersupply.

== Insert Figure 3 About Here ==

Contamination is substantial in most settings and is much higher when second-degree contamination is also considered. For example, with a sample of 500 ego networks per condition, contamination ranges from 2% to 45% under first-degree contamination (Figure 3(A)), compared with 34% to 99% under both first- and second-degree contamination (Figure 3(B)). Contamination is also influenced by sample size, population network size, and population network density. As sample size increases, contamination rate rises. For example, in the case of Twitch-Gamer, contamination increases from 4% with a sample size of 100 to 39% with a sample size of 1,000 under first-degree contamination. In contrast, contamination tends to decrease with population network size. For example, contamination rate in PowerLaw-Cluster-Small (10,000 nodes) is at least twice that in PowerLaw-Cluster-Base (100,000 nodes). Finally, contamination increases with population network

density. Contamination rates in PowerLaw-Cluster-Base are about 4–5 times those in PowerLaw-Cluster-Sparse.

Underrepresentation. We compare the characteristics of excluded egos with those of the population. Table 2 reports the degree, clustering coefficient, and eigenvector centrality of excluded egos under first-degree contamination, averaged over 100 simulations with 500 ego networks per simulation. Across all six population networks, excluded egos differ substantially from the population. They tend to have more connections (i.e., higher degree) and occupy more central positions in the network (i.e., higher eigenvector centrality). This is as expected because highly connected egos are more likely to link to other sampled egos and be removed.

== Insert Table 2 About Here ==

To assess underrepresentation directly, we again sample n ego networks per condition for n from 100 to 1000 and compute the percentage difference in average degree between each post-exclusion sample and the population. Figure 4 reports these differences under first-degree and first- and second-degree contamination. These percentage differences capture the severity of underrepresentation.

== Insert Figure 4 About Here ==

We observe consistent patterns in how the four factors influence underrepresentation, analogous to their effects on undersupply. First, the percentage differences are always negative, confirming that post-exclusion samples systematically retain lower-degree nodes. Second, underrepresentation is much more severe when both first- and second-degree contamination are removed. For example, with a sample of 500 ego networks, degree differences range from -4% to -57% under first-degree contamination (Figure 4(A)), compared with -22% to -98% under both criteria (Figure 4(B)). Third, underrepresentation increases with sample size. For Twitch-Gamer, the degree gap increases from -27% to -65% when sample size increases from 100 to 1,000 under first-degree contamination. Fourth,

underrepresentation tends to decrease with population network size and increases with population network density, mirroring the patterns observed for contamination rates and undersupply.

In summary, underrepresentation and undersupply are substantial when the standard exclusion approach is used to remove contaminated ego networks. These problems become more pronounced as sample size increases, population network size decreases, and population network density increases. Moreover, when both first- and second-degree contamination must be controlled, the majority of egos may be disqualified, rendering the exclusion approach impractical in many realistic settings.

4.3 Sample Representativeness: Proposed Method vs. Exclusion Approach

We compare the representativeness of samples generated by our method with that of post-exclusion samples across the six population networks. For the exclusion approach, we begin with 500 ego networks per condition and remove egos subject to first-degree contamination. For representative sampling, we generate samples matching the resulting post-exclusion sample size. Each procedure is repeated 100 times.

We evaluate representative samples and post-exclusion samples on their representativeness on three types of variables: optimization variables (degree and clustering coefficient), related network attributes (eigenvector centrality and neighbor degree), and other individual characteristics (demographics in real networks; simulated variables Z1–Z3 in simulated networks). Sample representativeness is evaluated using Kolmogorov-Smirnov (KS) distance of these variables between sample and population distributions. A smaller KS value indicates closer alignment between sample and population for that specific variable. Figure 5 summarizes the results.

== Insert Figure 5 About Here ==

Across all six networks, representative samples consistently achieve smaller KS distances than post-exclusion samples on all evaluated variables. This is because our method explicitly minimizes the distance on the optimization variables, which also improves alignment on correlated attributes

while leaving uncorrelated variables unaffected. These findings are supported by both theoretical justifications (Web Appendix D) and simulations described below.

For the optimization variables (Figure 5(A)), representative samples are approximately 3–19 times more similar to the population than post-exclusion samples, with larger improvements in small (e.g., LastFM and Small) or dense (e.g., Twitch and Base) networks. For related network attributes (Figure 5(B)), representative samples are also more closely aligned with the population, owing to correlations between these attributes and the optimization variables.

For individual characteristics, we use available individual characteristics in the three real networks including a variety of categorical, continuous and binary variables, and find that representative samples are equally or more closely aligned with the population than post-exclusion samples. In the three simulated networks, we simulate three variables: $Z1$ and $Z2$, which are strongly and moderately correlated with degree, and $Z3$, which is uncorrelated.⁶ We find that representative sampling improves the alignment for variables correlated with optimization variables ($Z1$ and $Z2$), and performs similarly to the exclusion approach for uncorrelated variable ($Z3$) (Figure 5(C)), consistent with the theoretical justifications. Additional results are in Web Appendix E.

4.4 Boundary Conditions: Proposed Method vs. Exclusion Approach

We examine the boundary conditions of the two sampling methods by comparing the maximum sample size achievable while maintaining representativeness. Because representativeness typically decreases as sample size increases, we impose a fixed threshold: the KS distance between the sample and population degree distributions must be less than 0.1. Under this criterion, we search for the largest feasible sample size under each method. For representative sampling, we begin with 100 ego

⁶ $Z1$ and $Z2$ are strongly ($\rho = -0.7$) and moderately ($\rho = 0.3$) correlated with degree. The correlation was imposed using a Cholesky decomposition to induce the correlation between the standardized degree and a random normal variable, which was then rescaled to the range $[0,10]$ to ensure comparability across covariates. $Z3$ is a binary variable independently drawn from a Bernoulli distribution with probability 0.5.

networks per condition and increase the sample size in steps of 100 until the $KS < 0.1$ threshold is reached. For the exclusion approach, we also begin with a random sample of 100 ego networks per condition, remove contaminated egos, and increase the sample size by 100 each time to find the largest post-exclusion sample that satisfies $KS < 0.1$. We repeat the same procedure for all six population networks.

We find that the maximum feasible sample size achieved by our method is about 8 to 56 times larger than that of the exclusion approach for the three real-world networks and 3 to 6 times larger for the three simulated networks (Table 3), with larger improvement for denser or smaller networks. This is because our method proactively controls contamination through the distance constraint (Eq. 4), which can also be extended to control higher-order contamination.⁷ In contrast, the exclusion approach relies on post-hoc removal, resulting in unpredictable post-exclusion sample sizes. Counterintuitively, increasing the initial sample does not necessarily lead to a larger post-exclusion sample (see Web Appendix G for details).

== Insert Table 3 About Here ==

In sum, our method consistently generates larger and more representative ego network samples across a variety of network conditions. It is particularly useful for obtaining high-quality samples in challenging situations when large sample sizes are required, when the population network is small or dense, or when both first- and second-degree contamination must be controlled.

5. Improved Causal Inference with Representative Samples

We compare representative samples and post-exclusion samples on their performance in estimating both average and heterogeneous treatment effects using simulations. For representative samples, we

⁷ We sincerely thank the review team for the comment on second-degree contamination. We compare the representative sampling and the exclusion approach regarding sample representativeness and sample size when both first- and second-degree contamination are controlled. For results and discussions, see Web Appendix F.

use the standard difference-in-means (DIM) estimator. For post-exclusion samples, in addition to the DIM estimator, we consider two inverse probability of treatment weighting (IPTW) estimators — the Horvitz-Thompson (HT) and Hájek estimators (Horvitz and Thompson 1952, Hájek 1971, Aronow and Samii 2017).⁸ The motivation is that because the exclusion approach yields nonrepresentative samples, IPTW can be used to reweight individual outcomes for causal estimation.

Although IPTW estimators can correct the estimation bias due to nonrepresentative samples, they rely on certain conditions for unbiased estimation that may not hold in our context. Additionally, IPTW estimators tend to have larger variance than unweighted estimators such as DIM, which can reduce statistical power to detect treatment effects. We discuss these caveats in the simulation analyses below and in Web Appendix H.

We conduct all simulations using matched-size representative and post-exclusion samples for each of the six networks.⁹ As in Section 4, post-exclusion samples are obtained by starting with 500 egos per condition and removing those with first-degree contamination. Representative samples then match this remaining sample size. Each procedure is repeated 100 times per network. In both approaches, egos are uncontaminated and alters are retained only if linked to a single ego, ensuring clean attribution of indirect effects.

5.1 Response Model

We focus on the problem of identifying influential users for social marketing campaigns. In this context, companies need to understand both direct and indirect treatment effects. Consumers in social networks are different in both their responsiveness to treatment and their influence over others (Aral and Dhillon 2018). To capture this heterogeneity, we specify a response model in which egos’

⁸ We sincerely thank the review team for suggesting these alternative benchmarks.

⁹ Replication code is available at <https://github.com/yanyanli-mkt/representative-ego-network-sampling>.

treatment effects vary with their degree, and alters' outcomes depend on the ego generating the spillover (e.g., a high-influence vs. a low-influence ego).

Specifically, let ego i receive a random treatment $D_i \in \{0,1\}$. The outcomes of egos i and alter $j(i)$, who is connected to ego i , are given by:

$$\begin{aligned} Y_i &= \alpha + \beta \mathbf{x}_i + \tau_i D_i + \xi_i + \varepsilon_i, \\ \tau_i &= \lambda K_i + v_i, \end{aligned} \tag{7}$$

$$\begin{aligned} Y_{j(i)} &= \alpha + \beta \mathbf{x}_{j(i)} + \gamma_{j(i)} D_i + \xi_i + \varepsilon_{j(i)}, \\ \gamma_{j(i)} &= \theta \tau_i. \end{aligned} \tag{8}$$

The direct treatment effect for ego i , denoted by τ_i , is heterogeneous and depends on the ego's degree K_i and a stochastic component $v_i \sim N(0, 0.5)$. The ego passes along a fraction θ of the direct treatment effect to connected alters. Accordingly, the indirect treatment effect on alter $j(i)$, denoted by $\gamma_{j(i)}$, is ego-specific and proportional to τ_i . Outcomes within the same ego network may be correlated due to homophily or unobserved common shocks. To capture this dependence, we include an ego-network-specific component $\xi_i \sim N(0, \sigma_\xi)$ in both the ego's and alter's outcomes. Lastly, ε_i and $\varepsilon_{j(i)}$ are idiosyncratic error terms distributed as $N(0, \sigma_\varepsilon)$.

To form the baseline outcome $Y_i(D_i = 0)$, we include individual-level covariates $\mathbf{x}_i'$. Across the six networks, these covariates include individual degree K_i and other observed characteristics in the real networks, and the three simulated variables $Z1$, $Z2$ and $Z3$ in the simulated networks.¹⁰ We log-transform degree and other highly skewed variables and set the coefficients $\beta = [0.5, -0.5, 0.5, -0.5]$, the intercept $\alpha = 1$, $\sigma_\xi = 1$ and $\sigma_\varepsilon = 1$.

5.2 Average Direct and Indirect Treatment Effects Estimation

¹⁰ Web Appendix I shows the user characteristics included in the response model for each of the six population networks.

We first focus on the average direct treatment effect (ADTE, $\tau = \bar{\tau}_i$), which captures whether an intervention can significantly change the behavior of targeted individuals and by how much on average. The ADTE is estimated by the difference in means between outcomes of egos in the treatment condition S_T and those in the control condition S_C ,

$$\hat{\tau} = \frac{1}{n_T} \sum_{i \in S_T} Y_i - \frac{1}{n_C} \sum_{i \in S_C} Y_i. \quad (9)$$

Next, the average indirect treatment effect (AITE, $\gamma = \bar{\gamma}_{j(i)}$) captures the average spillover from treated egos to connected alters. We first compute the average alter outcome within each ego network and use the mean difference of this quantity between treatment and control conditions to estimate the AITE,

$$\hat{\gamma} = \frac{1}{n_T} \sum_{i \in S_T} \bar{Y}_{j(i)} - \frac{1}{n_C} \sum_{i \in S_C} \bar{Y}_{j(i)}, \quad (10)$$

where $\bar{Y}_{j(i)} = \frac{1}{K_i} \sum_{j \in A(i)} Y_j$ is the average outcome of all alters of ego i and K_i is ego i 's degree.

Clean identification of the ADTE requires comparing egos in the “isolated direct exposure” condition (treated and not connected to other sampled egos) with those in the “no exposure” condition (untreated and not connected to other egos). Likewise, clean identification of the AITE requires comparing alters connected only to a treated ego with those connected only to a control ego. Because representative samples are constructed to satisfy these uncontaminated exposure conditions, we directly apply the difference-in-means (DIM) estimator described above. For post-exclusion samples, in addition to DIM, we apply the Horvitz-Thompson (HT) and Hájek IPTW estimators, which reweight each observed outcome Y_i or $Y_{j(i)}$ by the inverse probability of being observed in the corresponding exposure condition. The exposure probabilities are computed via Monte Carlo simulation following Aronow and Samii (2017). See Web Appendix H for formal definitions and detailed discussion.

A key caveat is that IPTW is unbiased only when exposure conditions uniquely determine potential outcomes (the consistency condition in Aronow and Samii 2017). This assumption holds for egos but may not for alters. For instance, an alter connected to a treated ego may have different outcomes depending on which ego generates the spillover (e.g., high- vs. low-influence egos). In our response model, an alter’s outcome is ego-specific and depends on the ego’s treatment effect τ_i , violating this requirement and making IPTW unreliable for AITE. Additionally, IPTW increases variance when exposure probabilities are small or highly unequal, which is common in network settings where nodes’ exposure probabilities often vary with network attributes (e.g., degree or clustering coefficient), reducing statistical power even when the consistency condition holds.

We evaluate estimator performance using mean absolute error (MAE), root mean squared error (RMSE), statistical power (the proportion of simulations rejecting the null hypothesis), and coverage (the proportion of confidence intervals containing the true value). To examine how performance varies with effect size, we vary the direct effect parameter λ from 0 to 0.1 in increments of 0.01, holding the spillover parameter θ fixed at 0.5. We repeat this exercise for all six population networks. For illustration, Figure 6 reports results for the PowerLaw-Cluster-Base network, which reflects an average case relative to the three real-world networks. Full results are provided in Web Appendix I.

== Insert Figure 6 About Here ==

The results reveal several consistent patterns for both ADTE and AITE. First, representative samples produce greater accuracy and stability than post-exclusion samples. The DIM estimator on representative samples achieves lower MAE and RMSE for both ADTE and AITE than any estimator on post-exclusion samples (top rows of Figures 6(A) and 6(B)). Second, representative samples provide more reliable inference, delivering both higher power and more accurate confidence-interval coverage. By contrast, for post-exclusion samples, the DIM estimator achieves similar power but exhibits poor coverage, indicating bias, whereas IPTW estimators improve coverage by correcting

for bias but at the cost of lower power due to higher variance (bottom rows of Figures 6(A) and 6(B)). Third, representative samples produce the most stable and accurate AITE estimates. In contrast, post-exclusion samples exhibit higher MAE and RMSE and lower power and coverage across all three estimators. Given the violation of consistency condition, IPTW estimators are not suitable for AITE in our setting and may introduce additional bias relative to DIM.

Overall, representative sampling improves the accuracy and reliability of both ADTE and AITE estimation, producing lower MAE and RMSE and higher power and coverage. By reducing bias without increasing variance, it enhances the efficiency of causal inference in network experiments using peer encouragement designs.

5.3 Heterogeneous Treatment Effect Estimation

Companies are also interested in identifying subpopulations that are more sensitive to the treatment or more influential over others in targeted campaigns. We illustrate how representative samples improve estimation of heterogeneous treatment effects across degree-based subgroups.

We estimate the conditional average direct treatment effect (CADTE) for all six networks, fixing the direct-effect parameter at $\lambda = 0.05$. For illustration, we report results for the simulated PowerLaw-Cluster-Base network.¹¹ Figure 7 compares CADTE estimates from representative samples using the DIM estimator with those from post-exclusion samples using DIM, HT and Hájek estimators.

== Insert Figure 7 About Here ==

Consistent with the ADTE and AITE results, representative samples produce CADTE estimates closer to the true effects and more stable across all degree subgroups, as reflected in lower MAE and RMSE. The advantage is especially pronounced for higher-degree subgroups. This is because high-degree nodes are more likely to be removed under the exclusion approach (Table 2), leaving too few

¹¹ Results for the remaining networks are available upon request.

high-degree egos in post-exclusion samples to reliably estimate subgroup treatment effects. Additional simulations confirm this explanation: the exclusion approach yields substantially fewer qualified ego networks in high-degree subgroups than representative sampling, with the gap widening substantially for higher-degree groups (see Web Appendix J for details).

In conclusion, our benchmark analysis demonstrates that representative sampling produces more accurate and reliable estimates of both average and heterogeneous treatment effects, achieving lower MAE and RMSE and higher power and coverage. Unlike IPTW estimators, representative samples reduce bias without increasing variance. The advantages are especially pronounced for high-degree subgroups, which are systematically underrepresented under the exclusion approach.

6. Conclusion

Peer encouragement designs are widely used in network experiments for evaluating targeted marketing campaigns; however, it remains challenging to obtain samples that are both representative and uncontaminated for estimating direct treatment effects on targeted individuals and indirect treatment effects on their connections. The common practice is to draw random samples and then remove contaminated units from analysis; however, this approach often leads to underrepresentation and undersupply. These problems become more pronounced when required sample sizes are large, when population networks are small or dense, or when both first- and second-degree contamination must be controlled.

We propose a representative ego network sampling method that directly addresses these limitations. Our approach uses the Metropolis-Hastings algorithm to generate samples whose joint distribution of key network attributes converges to that of the population. Rather than removing contaminated nodes post-hoc, our method proactively embeds a distance constraint on the sampled egos, generating samples that are both representative and uncontaminated.

Across simulated and real-world networks, we demonstrate that our method outperforms the exclusion approach. Representative samples better represent the population not only on the optimization variables (degree and clustering coefficient) but also on other network attributes or individual characteristics correlated with them. For variables independent of the optimization variables, representative and post-exclusion samples perform similarly, demonstrating that our method does not distort unrelated characteristics.

The advantages in sample representativeness translate directly into improvements in causal inference. Representative samples produce more accurate and stable estimates of both direct and indirect average treatment effects, with lower MAE and RMSE and higher power and coverage. For heterogeneous treatment effects, estimates of the conditional average direct treatment effects are more precise, especially for high-degree subgroups that are systematically underrepresented under the exclusion approach. A key advantage of our method is that it does not require IPTW correction and therefore avoids the variance inflation and the consistency condition limitations that constrain IPTW estimators in network settings.

While our analyses focus on controlling first-degree contamination as done in previous research using the exclusion approach, our method can be easily extended to control both first- and second-degree contamination by adjusting the distance threshold parameter $h = 3$. We have demonstrated that the exclusion approach becomes impractical under such strict criteria, often resulting in extremely small sample sizes. Representative sampling maintains both representativeness and adequate sample size, highlighting the flexibility and robustness of our method, especially in small or dense networks.

In summary, our method provides a practical and effective tool for researchers and practitioners designing network experiments. By generating representative and uncontaminated ego network samples, our method enables more accurate estimation of average and heterogeneous treatment effects, improving the external validity and managerial usefulness of peer encouragement designs.

These advantages are especially important when treatments are costly, targeted populations are small or densely connected, or precise estimates are needed to identify the most responsive or influential individuals.

Our research can be extended in several ways, offering opportunities for future research. First, we consider heterogeneity among egos in their responses to the treatment and the social influence they exert on their connections. A natural extension is to examine heterogeneity among alters to better understand how indirect treatment effects or social influences vary across alter subpopulations differing in demographics or network attributes (e.g., Aral and Dhillon 2018). This can be achieved by adding a second-stage representative sampling step for alters using an adapted Metropolis-Hastings procedure, ensuring that sampled alters are also representative of the population.

Second, while this study focuses on peer encouragement designs, the sampling framework can be adapted for other types of network experiments, such as the mechanism design experiments that compare alternative modes of social influence rather than presence versus absence of treatment (e.g., Aral and Walker 2011). Causal inference in such settings becomes difficult when an alter is connected to multiple egos assigned to different mechanisms because the alter's observed response cannot be attributed cleanly to any single mechanism. Our method can address this challenge by modifying the objective function and distance constraint to ensure that each alter is connected to only one ego and that alters within each condition are representative of the population on the characteristics relevant to the analysis.

Funding and Competing Interests

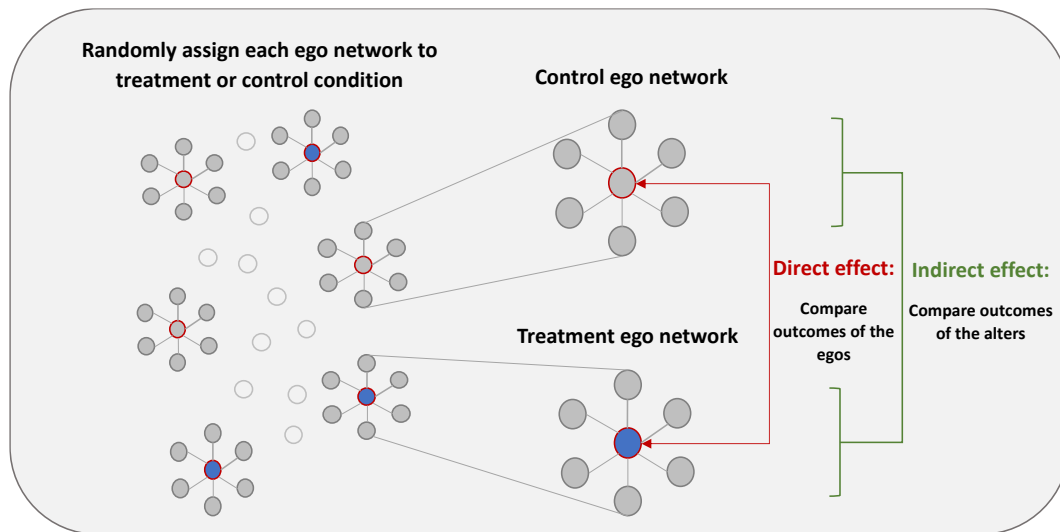
All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no funding to report.

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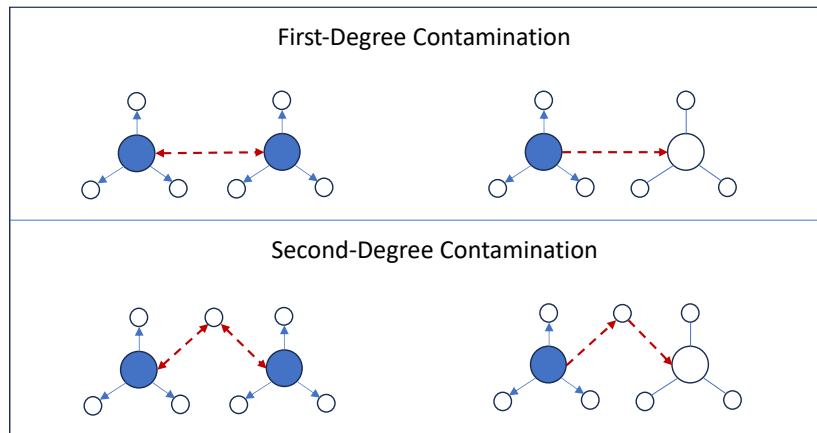
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Figure 1. Illustration of the Peer Encouragement Design and Estimation of Treatment Effects



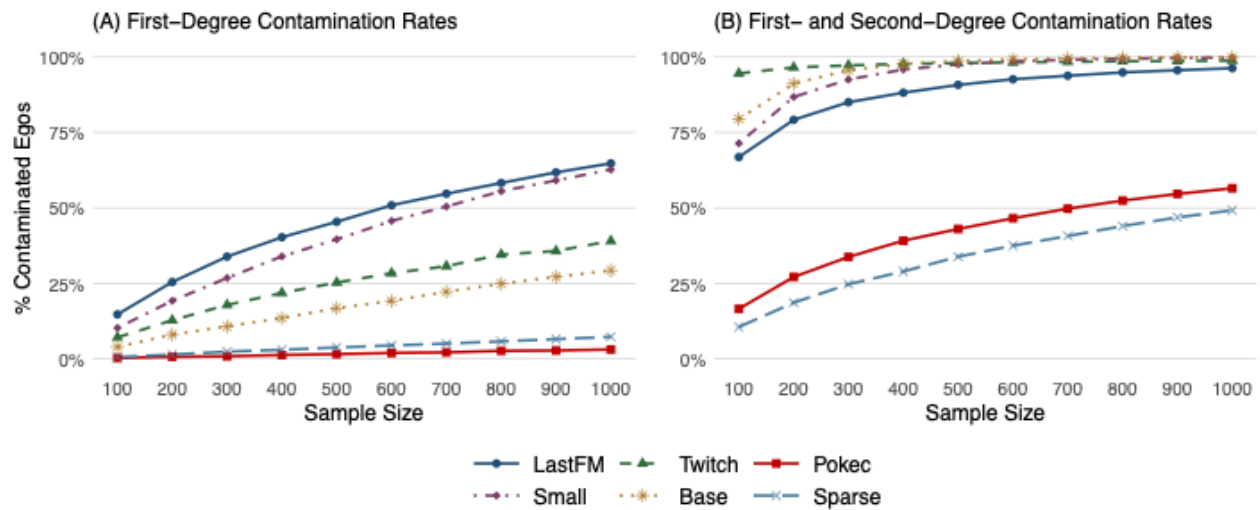
Note. An ego network consists of an ego (shown as an outlined node) and the alters connected to that ego. In a peer encouragement design, ego networks are randomly assigned to treatment or control, and only egos in the treatment condition receive the intervention (shown as filled nodes). The direct treatment effect is estimated by comparing the average outcomes of egos in the treatment and control conditions. The indirect treatment effect is estimated by comparing the average outcomes of alters connected to treated egos with those connected to control egos.

Figure 2. First- and Second-Degree Contamination



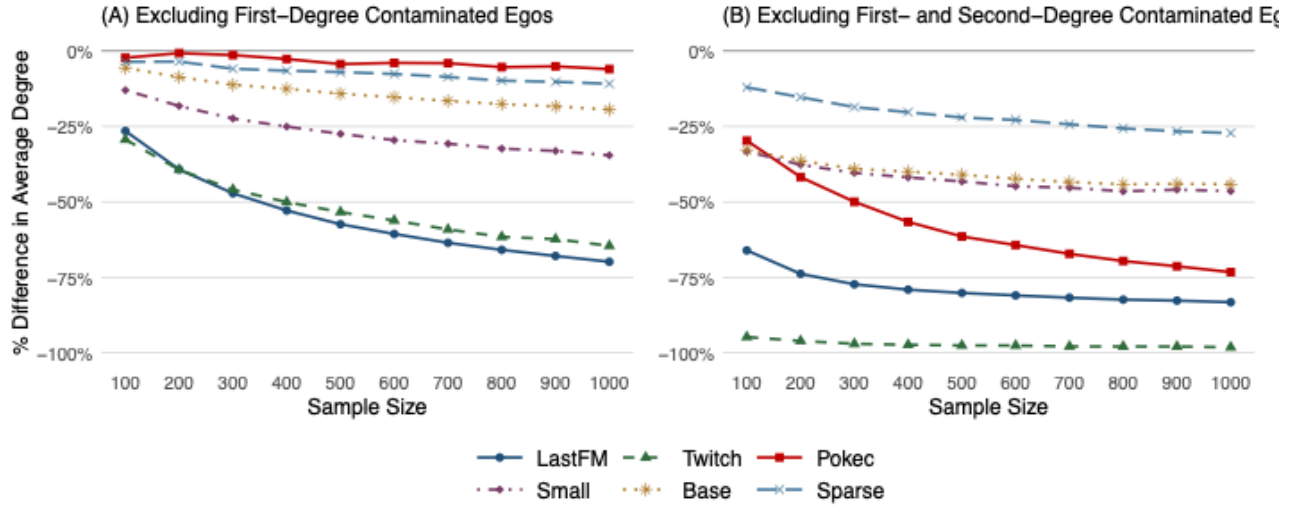
Note. Nodes are distinguished by treatment status, with treated nodes shown as filled circles and untreated nodes shown as unfilled circles. Solid arrows indicate influence links. Dashed arrows indicate potential paths of treatment contamination, illustrating first-degree contamination in the top panel and second-degree contamination in the bottom panel.

Figure 3. Contamination Rates by Sample Size Across Population Networks



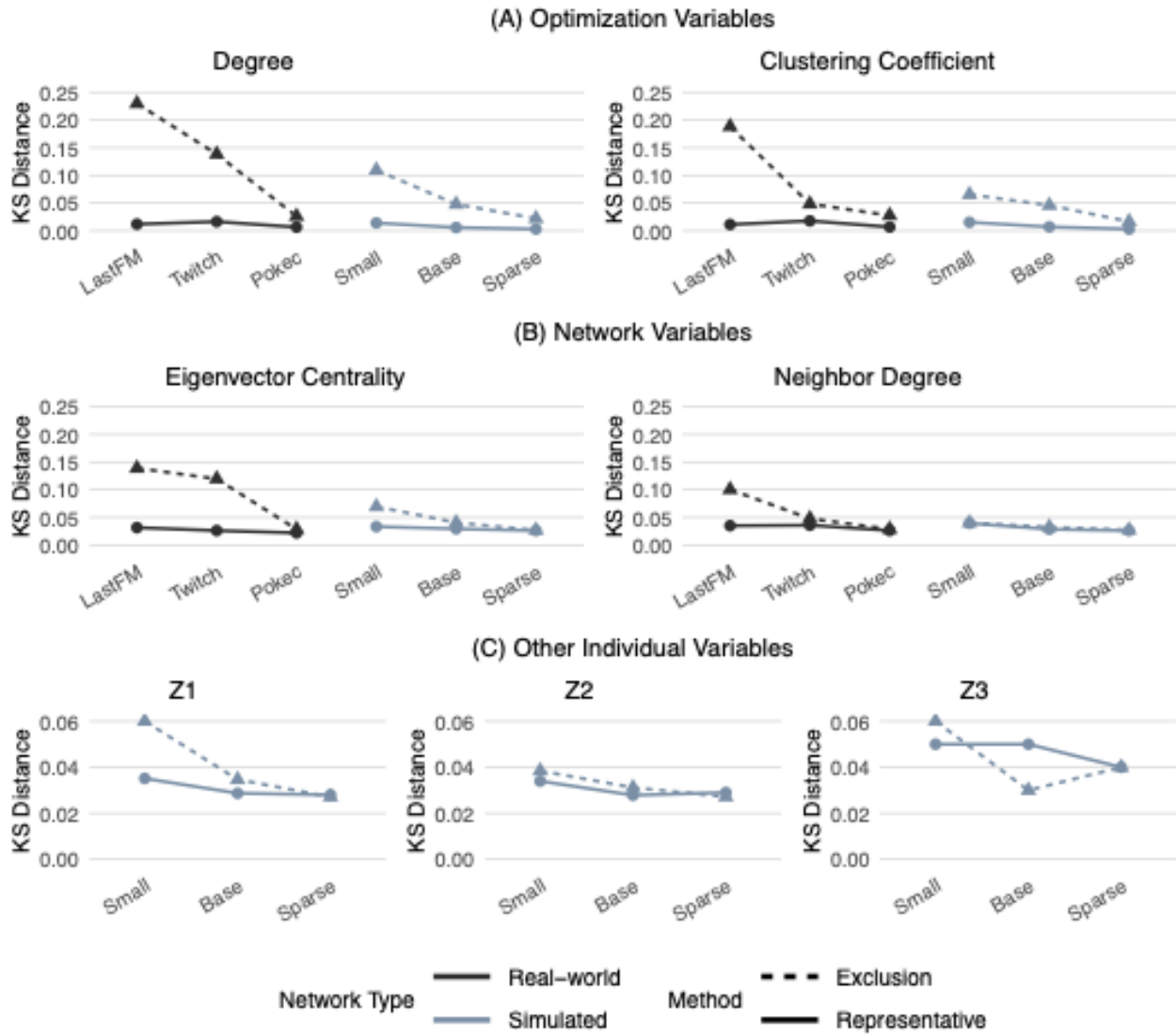
Note. This figure presents contamination rates (vertical axis) by sample size (horizontal axis) for the six population networks, with one line for each network. The contamination rate is the percentage of sampled egos that are contaminated.

Figure 4. Percentage Difference in Average Degree Between the Sample and the Population



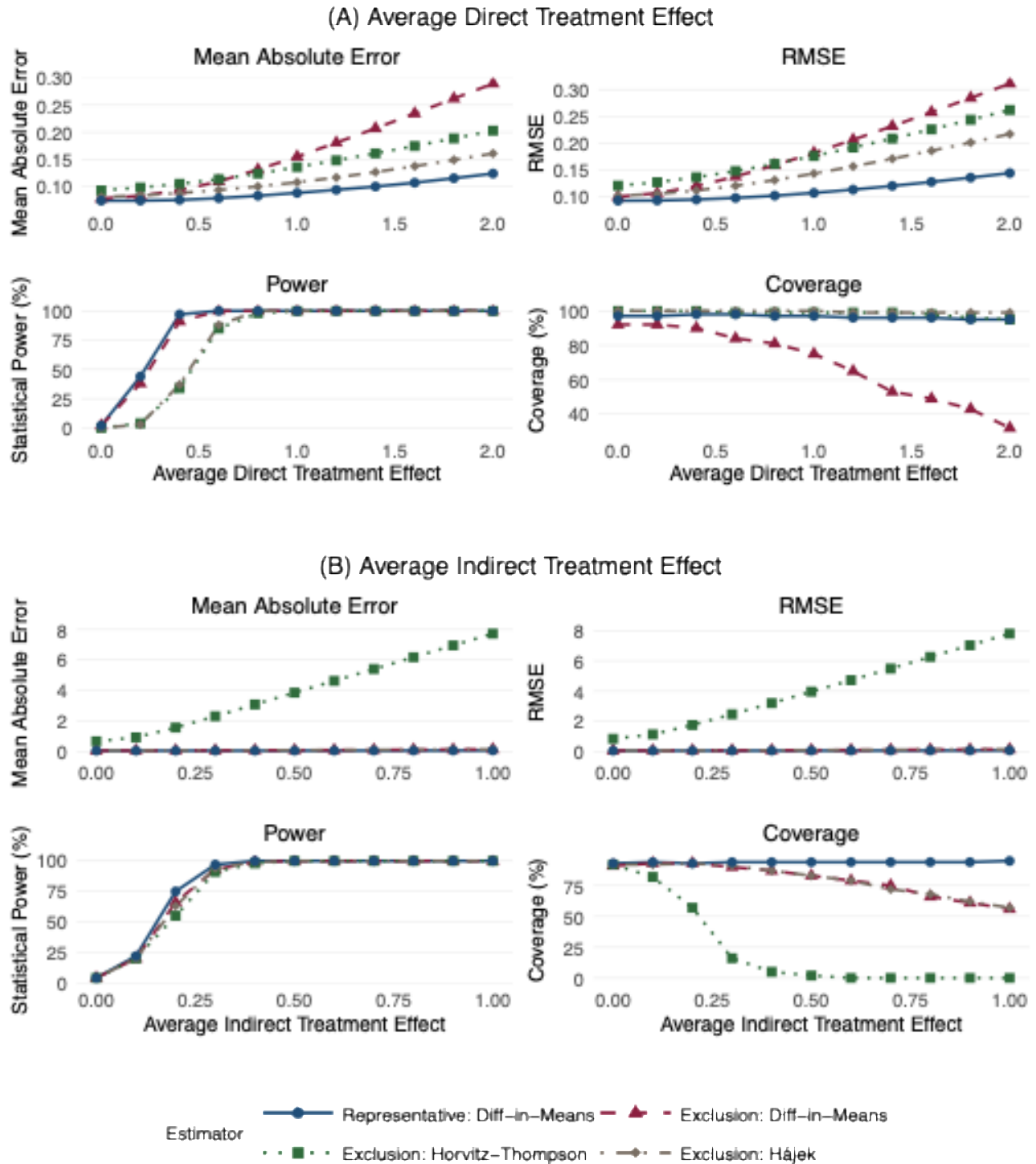
Note. This figure presents the percentage difference between the sample's average degree and the population average degree (vertical axis) by sample size (horizontal axis) for the six population networks, with one line for each network.

Figure 5. Representativeness of the Representative and Post-Exclusion Samples



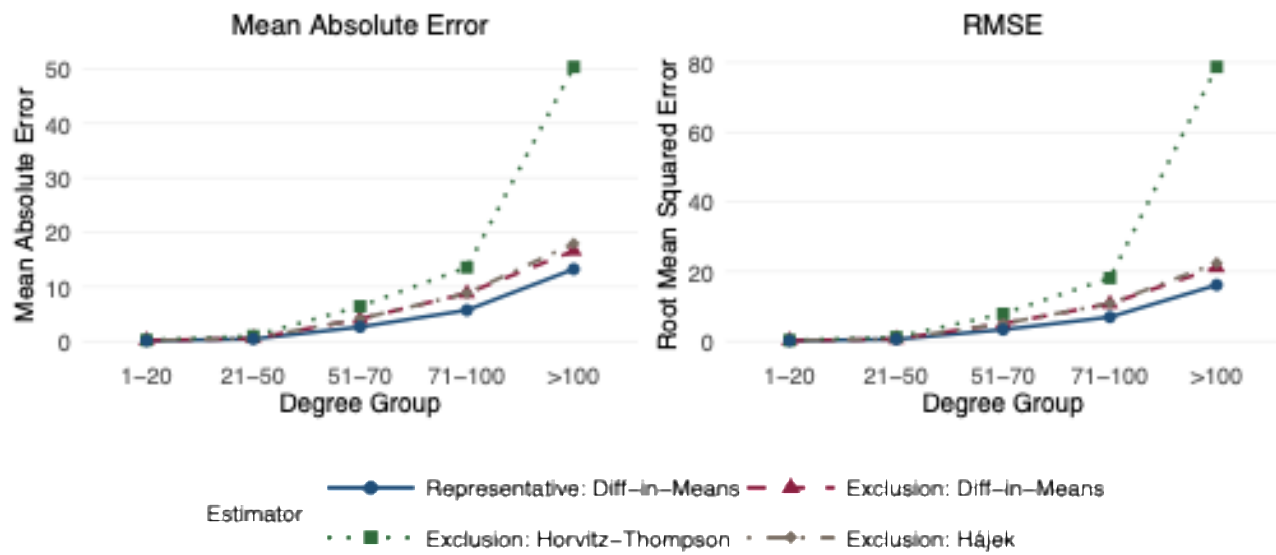
Note. This figure reports the Kolmogorov–Smirnov (KS) distances for each variable under the representative sampling method and the exclusion approach across the population networks. Figure 5(C) focuses on the three simulated variables (Z1, Z2, and Z3) in the three simulated networks to facilitate comparison.

Figure 6. Estimator Performance for the ADTE and AITE Across Effect Sizes



Note. This figure reports the mean absolute error (MAE), root mean squared error (RMSE), power (the proportion of simulations rejecting the null hypothesis), and coverage (the proportion of confidence intervals containing the true value) for estimating the average direct treatment effect (ADTE) and average indirect treatment effect (AITE), respectively. The four estimators are the difference-in-means, Horvitz–Thompson, and Hájek estimators applied to post-exclusion samples, and the difference-in-means estimator applied to representative samples. Results are based on the simulated PowerLaw-Cluster-Base network. The effect size varies from 0 to 2 for ADTE and from 0 to 1 for AITE.

Figure 7. Estimator Performance for the CADTE Across Degree Groups



Note. This figure reports the mean absolute error and root mean squared error for estimating the conditional average direct treatment effect across different degree groups using the four estimators. Results are based on the simulated PowerLaw-Cluster-Base network.

Table 1. Summary of the Population Networks

	Real-World Networks			Simulated Networks		
	LastFM	Twitch	Pokec	Small	Base	Sparse
Population size	7,624	168,114	1,632,803	10,000	100,000	100,000
Number of edges	27,806	6,797,557	30,622,564	29,988	998,811	199,996
Average degree	7.294	80.868	27.317	5.998	19.996	4.000
Average clustering coefficient	0.220	0.160	0.109	0.158	0.166	0.075
Average eigenvector centrality	0.002	0.001	0.0001	0.005	0.001	0.0005
Density	0.001	0.0005	0.00002	0.0006	0.0002	0.00004
Average neighbor degree	23.924	2215.552	86.297	50.103	120.765	21.855

Note. This table reports the summary statistics for network characteristics for the six population networks.

Table 2. Network Characteristics of Population Nodes and Egos Excluded for First-Degree Contamination

	LastFM		Twitch		Pokec	
	Population	Excluded	Population	Excluded	Population	Excluded
Average degree	7.294	12.247	80.868	203.350	27.317	87.246
Average clustering coefficient	0.220	0.256	0.160	0.130	0.109	0.083
Average eigenvector centrality	0.002	0.004	0.001	0.002	0.0001	0.0006
	Small		Base		Sparse	
	Population	Excluded	Population	Excluded	Population	Excluded
Average degree	5.998	8.516	19.996	34.510	4.000	11.304
Average clustering coefficient	0.158	0.138	0.166	0.140	0.075	0.053
Average eigenvector centrality	0.005	0.006	0.001	0.002	0.0005	0.0014

Note. This table compares the characteristics of population nodes and egos excluded because of first-degree contamination across the six population networks. In each simulation, we sample 500 ego networks per condition. Results are averaged across 100 simulations.

Table 3. Maximum Feasible Sample Size Per Condition with First-Degree Contamination Controlled and $KS < 0.1$

Population network	Representative Sampling	Exclusion Approach
LastFM	900	85
Twitch	11,800	211
Pokec	50,000	6,426
Small	1,300	266
Base	5,100	862
Sparse	13,500	4,144

Note. The results are averaged across 100 simulations. For the three smaller or denser networks (LastFM, Twitch, and Small), we increase sample sizes from 50 to 500 per condition in steps of 50 to locate the boundary more precisely.